

THE FERRERS BOUND FOR SPANNING TREES IN BIPARTITE GRAPHS

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ABSTRACT. We prove Ehrenborg’s conjecture that every connected bipartite graph G with parts of size m and n has at most $\frac{1}{mn} \prod_{v \in V(G)} \deg(v)$ spanning trees, and that equality holds if and only if G is a Ferrers graph.

1. INTRODUCTION

Given a bipartite graph $G = (X \sqcup Y, E)$, its *Ferrers invariant* is defined by

$$F(G) := \frac{1}{|X||Y|} \prod_{v \in V(G)} \deg(v).$$

Ehrenborg conjectured in 2006 that the number $\tau(G)$ of spanning trees in G is at most $F(G)$ for every bipartite G (see [17]). In this paper, we prove this conjecture and characterize the equality case. To avoid trivial issues, we will assume that all graphs are connected in what follows.

The origin of the problem lies in the class of *Ferrers graphs*, studied by Ehrenborg and van Willigenburg [6]. A connected bipartite graph is Ferrers if the neighborhoods of the vertices on one side can be linearly ordered by inclusion.

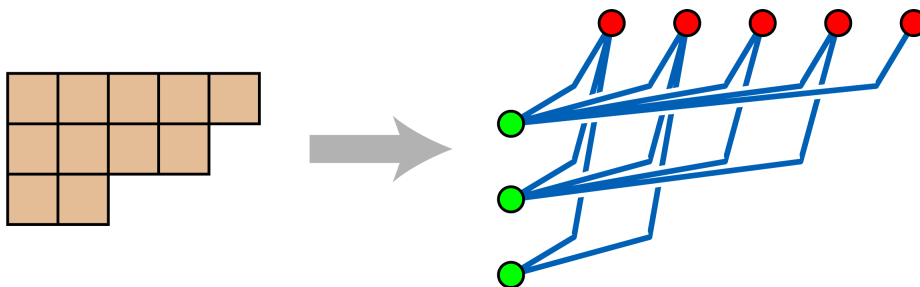


FIGURE 1. The edges of a Ferrers graph are in direct correspondence with the boxes in a Ferrers diagram. (Reproduced from David Eppstein, “Four open from IPAM” [7], 2009, CC BY 4.0 license.)

Ehrenborg and van Willigenburg proved that Ferrers graphs satisfy the exact product formula

$$\tau(G) = F(G),$$

and in fact established a weighted refinement, using ideas from the theory of electrical networks [6]. The Ferrers equality case has since been revisited from several directions: Burns gave bijective proofs [4], Bapat gave another proof via resistance distance [1], Volkova

obtained a proof by inductively computing the determinant of a matrix [19], and Go, Khwa, Luo, and Stamps later rederived the Ferrers formula using triangular rank-one perturbations of Laplacian matrices [11].

The general conjecture has nevertheless resisted earlier attacks. Garrett and Klee [9] reformulated the conjecture in terms of the nonnegativity of a certain homogeneous polynomial, and used this reformulation to verify the conjecture when one part has size at most 5. They also proved separately that adding a pendant vertex preserves the conjecture, which yields the tree case. See also Slone's survey [17] and Bapat's exposition [1]. Koo's senior thesis [15] established further special cases, including even cycles and sufficiently dense bipartite graphs with a cut vertex of degree 2, and also showed that the operation of connecting two graphs by a new edge preserves Ferrers-goodness (that is, preserves the bound $\tau(G) \leq F(G)$). Slone [17] also proved that gluing Ferrers-good graphs along a vertex preserves Ferrers-goodness, and consequently that any minimal counterexample must be 2-connected; the same paper recorded computational verification up to 13 vertices and proposed a stronger majorization-based statement that was shown there to be false. Later work of Slone [18] extended the computational verification from 13 to 17 vertices. In 2020, Volkova proved the conjecture for one-side regular bipartite graphs [19].

At the same time, several complementary reformulations and weaker bounds were developed. Bozkurt proved the general bipartite estimate

$$\tau(G) \leq \frac{1}{|E(G)|} \prod_{v \in V(G)} \deg(v),$$

with equality exactly for complete bipartite graphs [3]; this is weaker than the Ferrers bound by a factor of 1 over the bipartite density $|E(G)|/(|X||Y|)$. Spectral and majorization reformulations were recorded by Slone and Bapat [17, 1], and more recently Cherkashin and Prozorov showed that a natural polynomial version of Ehrenborg's conjecture is equivalent to the original inequality [5, Theorem 2]. Consequently, the following is a corollary of our result:

Corollary. *Let $G = (X \sqcup Y, E)$ be a bipartite graph. Define*

$$P_G((z_v)_{v \in V(G)}) := \sum_T \prod_{v \in V(G)} z_v^{\deg_T(v)-1},$$

where the sum ranges over all spanning trees T of G (so P_G is the zero polynomial if G is disconnected). Then, for every choice of nonnegative reals z_v ($v \in V(G)$), we have

$$P_G((z_v)_{v \in V(G)}) \left(\sum_{x \in X} z_x \right) \left(\sum_{y \in Y} z_y \right) \leq \prod_{v \in V(G)} \left(\sum_{u \in N_G(v)} z_u \right).$$

The proof of our result is linear-algebraic, following the approach of Klee and Stamps [13, 14]: We first add a rank-one matrix to the graph Laplacian, and apply Kirchhoff's theorem to evaluate the determinant of this perturbed matrix. Schur complementation then reduces the problem to the positive definite matrix M on the X -side. Our key contribution is recognizing that this matrix is a sum of orthogonal projections indexed by the neighborhoods of the vertices of Y . A trace identity compares these projections with prefix sums of the degree sequence. Ky Fan's maximum principle then gives a majorization of the degree sequence by the eigenvalues of the reduced matrix, and strict Schur-concavity yields the determinant bound. The same trace identity characterizes equality through a nonnegative *overlap defect*.

Now let $G = (X \sqcup Y, E)$ be a bipartite graph with $|X| = m$ and $|Y| = n$. Following [6], we call G a *Ferrers graph* if, after ordering

$$X = \{x_1, \dots, x_m\}, \quad Y = \{y_1, \dots, y_n\},$$

there exist integers

$$m = t_1 \geq t_2 \geq \dots \geq t_n \geq 1$$

such that

$$N(y_j) = \{x_1, \dots, x_{t_j}\} \quad (1 \leq j \leq n).$$

Equivalently, the neighborhoods on one side of the bipartition are linearly ordered by inclusion. In terms of the biadjacency matrix, this means that after a suitable ordering, every column consists of a block of 1's followed by 0's, and the column heights are weakly decreasing; in other words, the pattern of 1's forms a Ferrers diagram.

The main result of this paper is the following:

Theorem 1. *Let $G = (X \sqcup Y, E)$ be a connected bipartite graph with $|X| = m$ and $|Y| = n$. Then*

$$\tau(G) \leq \frac{1}{mn} \prod_{v \in V(G)} \deg(v).$$

Moreover, equality holds if and only if G is a Ferrers graph.

2. DETERMINANT REDUCTION

Standard ingredients. We use Kirchhoff's matrix-tree theorem from algebraic graph theory [10, Section 13.2], the Schur complement determinant formula and the matrix determinant lemma from matrix analysis [12, Section 0.8.5], and Ky Fan's maximum principle [2, Exercise II.1.13]; see also [8]. We also use the strict form of Karamata's inequality [16, Section 3C].

Theorem 2 (Kirchhoff's matrix-tree theorem). *Let H be a graph with Laplacian matrix $L(H)$. Deleting the same row and column from $L(H)$ produces a submatrix whose determinant is independent of the deleted vertex and equals $\tau(H)$.*

Proposition 3 (Schur complement determinant formula). *Let U and Z be square matrices, and let V and W be compatible rectangular blocks. If Z is invertible, then*

$$\det \begin{pmatrix} U & V \\ W & Z \end{pmatrix} = \det(Z) \det(U - VZ^{-1}W).$$

Recall that for a square matrix S , the *adjugate* $\text{adj}(S)$ is the transpose of its cofactor matrix and satisfies

$$S \text{adj}(S) = \text{adj}(S)S = (\det S)I.$$

Lemma 4. *Let H be a connected graph on N vertices, let $L = L(H)$, and let $\mathbf{1}_N$ be the all-ones vector. Then, for every $z \in \mathbb{R}^N$,*

$$\det(L + zz^\top) = \tau(H)(\mathbf{1}_N^\top z)^2. \tag{1}$$

Proof. Connectedness gives $\ker L = \langle \mathbf{1}_N \rangle$. Since

$$L \text{adj}(L) = \text{adj}(L)L = 0,$$

every row and column of $\text{adj}(L)$ is constant. Hence

$$\text{adj}(L) = c \mathbf{1}_N \mathbf{1}_N^\top$$

for some scalar c . By Kirchhoff's theorem, each diagonal cofactor of L is $\tau(H)$, so $c = \tau(H)$. The matrix determinant lemma [12, Section 0.8.5] states that

$$\det(S + uv^\top) = \det S + v^\top \text{adj}(S)u.$$

Since $\det L = 0$, it follows that

$$\det(L + zz^\top) = z^\top \text{adj}(L)z = \tau(H)(\mathbf{1}_N^\top z)^2.$$

□

Definition 5. For $x, y \in \mathbb{R}^m$, write $x^\downarrow$ and $y^\downarrow$ for their coordinates rearranged in weakly decreasing order. We say that x *majorizes* y if

$$\sum_{i=1}^k x_i^\downarrow \geq \sum_{i=1}^k y_i^\downarrow \quad (1 \leq k < m)$$

and

$$\sum_{i=1}^m x_i = \sum_{i=1}^m y_i.$$

Proposition 6 (Strict Schur concavity). *If $x, y \in (0, \infty)^m$ and x majorizes y , then for every strictly concave function $\phi: (0, \infty) \rightarrow \mathbb{R}$,*

$$\sum_{i=1}^m \phi(x_i) \leq \sum_{i=1}^m \phi(y_i).$$

If, in addition, x and y are both written in weakly decreasing order, then equality holds if and only if $x_i = y_i$ for all i .

Proof. This is the standard strict form of Karamata's inequality, or equivalently the statement that $\sum_i \phi(x_i)$ is strictly Schur-concave when ϕ is strictly concave; see [16, Section 3C]. □

Proposition 7 (Ky Fan's maximum principle). *Let S be a real symmetric $m \times m$ matrix with eigenvalues*

$$\theta_1 \geq \theta_2 \geq \cdots \geq \theta_m.$$

If R is an orthogonal projection of rank k , then

$$\text{tr}(RS) \leq \sum_{i=1}^k \theta_i. \tag{2}$$

Equality is attained when R projects onto the span of eigenvectors corresponding to $\theta_1, \dots, \theta_k$.

Proof. Choose an orthonormal eigenbasis $v_1, \dots, v_m$ with $Sv_i = \theta_i v_i$, and put

$$c_i = \langle Rv_i, v_i \rangle = \|Rv_i\|^2.$$

Then $0 \leq c_i \leq 1$ and $\sum_i c_i = \text{tr } R = k$. Therefore

$$\begin{aligned} \text{tr}(RS) &= \sum_{i=1}^m \theta_i c_i \\ &= \sum_{i=1}^k \theta_i - \sum_{i=1}^k \theta_i (1 - c_i) + \sum_{i=k+1}^m \theta_i c_i \\ &\leq \sum_{i=1}^k \theta_i - \theta_k \sum_{i=1}^k (1 - c_i) + \theta_k \sum_{i=k+1}^m c_i = \sum_{i=1}^k \theta_i, \end{aligned}$$

since the last two sums are equal. Projection onto $\text{span}\{v_1, \dots, v_k\}$ gives equality. $\square$

Order the vertices so that those of X come first. The Laplacian of G has block form

$$L(G) = \begin{pmatrix} A & -B \\ -B^\top & C \end{pmatrix},$$

where $B \in \{0, 1\}^{m \times n}$ is the biadjacency matrix of G , and

$$A = \text{diag}(a_1, \dots, a_m), \quad C = \text{diag}(b_1, \dots, b_n),$$

with $a_i = \deg(x_i)$ and $b_j = \deg(y_j)$. Since G is connected, every b_j is positive, so C is invertible. Let

$$L_X := A - BC^{-1}B^\top, \quad J := \mathbf{1}_m \mathbf{1}_m^\top, \quad \Pi := \frac{1}{m} J,$$

and define

$$M := L_X + n\Pi = L_X + \frac{n}{m} J. \quad (3)$$

Proposition 8. *We have*

$$\tau(G) = \frac{\prod_{j=1}^n b_j}{mn} \det M. \quad (4)$$

Consequently, the inequality in Theorem 1 is equivalent to

$$\det M \leq \prod_{i=1}^m a_i. \quad (5)$$

Proof. Apply Lemma 4 to $L(G)$ with

$$z := \begin{pmatrix} \sqrt{n/m} \mathbf{1}_m \\ 0 \end{pmatrix} \in \mathbb{R}^{m+n}.$$

Since $\mathbf{1}_{m+n}^\top z = \sqrt{mn}$, we obtain

$$mn \tau(G) = \det \begin{pmatrix} A + n\Pi & -B \\ -B^\top & C \end{pmatrix}.$$

Taking the Schur complement of C gives

$$mn \tau(G) = \det(C) \det(A + n\Pi - BC^{-1}B^\top) = \left(\prod_{j=1}^n b_j \right) \det M,$$

which proves (4). Since all degrees are positive, the desired spanning-tree inequality is equivalent to (5). $\square$

3. PROJECTION TRACES

For each $y_j \in Y$, write

$$T_j := N(y_j) \subseteq X,$$

so that $b_j = |T_j|$. For every nonempty $T \subseteq X$, let $\mathbf{1}_T \in \mathbb{R}^m$ be its indicator vector and define

$$D_T := \text{diag}(\mathbf{1}_T), \quad J_T := \mathbf{1}_T \mathbf{1}_T^\top, \quad P_T := D_T - \frac{1}{|T|} J_T.$$

Also set

$$H_T := \left\{ u \in \mathbb{R}^m : \text{supp}(u) \subseteq T, \sum_{x_i \in T} u_i = 0 \right\}, \quad Q_T := P_T + \Pi.$$

Lemma 9. *The matrix P_T is the orthogonal projection onto H_T , and Q_T is the orthogonal projection onto*

$$U_T := H_T \oplus \langle \mathbf{1}_m \rangle.$$

In particular, Q_T has rank $|T|$.

Proof. For $u \in \mathbb{R}^m$, let

$$\bar{u}_T := \frac{1}{|T|} \sum_{x_i \in T} u_i.$$

Then

$$(P_T u)_i = \begin{cases} u_i - \bar{u}_T, & x_i \in T, \\ 0, & x_i \notin T. \end{cases}$$

Thus $P_T u \in H_T$, while for every $w \in H_T$,

$$\langle u - P_T u, w \rangle = \bar{u}_T \sum_{x_i \in T} w_i = 0.$$

Hence P_T is the orthogonal projection onto H_T . The matrix Π is the orthogonal projection onto $\langle \mathbf{1}_m \rangle$, and this line is orthogonal to H_T . Therefore $Q_T = P_T + \Pi$ is the orthogonal projection onto U_T . Finally, $\dim H_T = |T| - 1$, so $\text{rank}(Q_T) = |T|$. $\square$

Since the j th column of B is $\mathbf{1}_{T_j}$, we have

$$A = \sum_{j=1}^n D_{T_j}, \quad BC^{-1}B^\top = \sum_{j=1}^n \frac{1}{|T_j|} J_{T_j}.$$

Together with (3), this gives the projection decomposition

$$M = \sum_{j=1}^n Q_{T_j}. \tag{6}$$

In particular, M is positive semidefinite. Proposition 8 and connectedness give $\det M > 0$, so M is in fact positive definite.

For nonempty $I, T \subseteq X$, define the *overlap defect*

$$\varepsilon(I, T) := \frac{|I \setminus T| |T \setminus I|}{|I| |T|}.$$

Lemma 10. *For all nonempty $I, T \subseteq X$,*

$$\text{tr}(Q_I Q_T) = |I \cap T| + \varepsilon(I, T). \quad (7)$$

The defect is nonnegative and vanishes if and only if $I \subseteq T$ or $T \subseteq I$.

Proof. Write

$$p := |I|, \quad q := |T|, \quad r := |I \cap T|.$$

Because $P_I \Pi = \Pi P_T = 0$ and $\Pi^2 = \Pi$,

$$\text{tr}(Q_I Q_T) = \text{tr}(P_I P_T) + 1.$$

A direct trace calculation gives

$$\text{tr}(P_I P_T) = r - \frac{r}{p} - \frac{r}{q} + \frac{r^2}{pq}.$$

Therefore

$$\text{tr}(Q_I Q_T) = r + \frac{(p-r)(q-r)}{pq} = |I \cap T| + \varepsilon(I, T).$$

The final term is nonnegative and vanishes exactly when $p = r$ or $q = r$. $\square$

The overlap identity and the projection decomposition combine into the central trace formula of the proof.

Proposition 11. *For every nonempty $I \subseteq X$,*

$$\text{tr}(Q_I M) = \sum_{x_i \in I} a_i + \sum_{j=1}^n \varepsilon(I, T_j). \quad (8)$$

Proof. By (6) and Lemma 10,

$$\text{tr}(Q_I M) = \sum_{j=1}^n \text{tr}(Q_I Q_{T_j}) = \sum_{j=1}^n |I \cap T_j| + \sum_{j=1}^n \varepsilon(I, T_j).$$

The first sum counts the edges incident to vertices of I , and hence equals $\sum_{x_i \in I} a_i$. $\square$

4. PROOF OF THE THEOREM

Relabel the vertices of X so that

$$a_1 \geq a_2 \geq \cdots \geq a_m > 0,$$

and write $[k] = \{x_1, \dots, x_k\}$. Let

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_m > 0$$

be the eigenvalues of M .

For each k , the matrix $Q_{[k]}$ is an orthogonal projection of rank k . Ky Fan's maximum principle and (8) therefore give

$$\sum_{i=1}^k \lambda_i \geq \text{tr}(Q_{[k]} M) = \sum_{i=1}^k a_i + \sum_{j=1}^n \varepsilon([k], T_j) \geq \sum_{i=1}^k a_i. \quad (9)$$

For $k = m$, one has $Q_{[m]} = I$ and every overlap defect vanishes, so

$$\sum_{i=1}^m \lambda_i = \operatorname{tr} M = \sum_{i=1}^m a_i.$$

Thus $(\lambda_1, \dots, \lambda_m)$ majorizes $(a_1, \dots, a_m)$. Applying Proposition 6 to the strictly concave function $\log$ gives

$$\log \det M = \sum_{i=1}^m \log \lambda_i \leq \sum_{i=1}^m \log a_i,$$

and hence

$$\det M \leq \prod_{i=1}^m a_i.$$

Proposition 8 proves the inequality in Theorem 1.

It remains to characterize equality. Suppose first that equality holds in Theorem 1. Then $\det M = \prod_i a_i$, so equality holds in the strict Schur-concavity inequality above. Since both vectors are weakly decreasing,

$$\lambda_i = a_i \quad (1 \leq i \leq m).$$

The two ends of (9) are consequently equal for every k , and therefore

$$\sum_{j=1}^n \varepsilon([k], T_j) = 0.$$

Every summand is nonnegative, so each T_j is comparable by inclusion with every initial segment $[k]$.

Fix T_j , and let t be the largest index for which $x_t \in T_j$. If $t = 1$, then $T_j = [1]$. If $t > 1$, then $T_j \not\subseteq [t-1]$, so comparability with $[t-1]$ implies $[t-1] \subseteq T_j$. Since $T_j \subseteq [t]$ and $x_t \in T_j$, it follows that $T_j = [t]$. Thus every neighborhood on the Y -side is an initial segment in the same ordering of X . Reordering the vertices of Y by decreasing neighborhood size shows that G is Ferrers.

Conversely, suppose that G is Ferrers, so after suitable orderings

$$T_j = [t_j] \quad \text{with} \quad m = t_1 \geq t_2 \geq \dots \geq t_n \geq 1.$$

The subspaces

$$\operatorname{im} Q_{[1]} \subset \operatorname{im} Q_{[2]} \subset \dots \subset \operatorname{im} Q_{[m]} = \mathbb{R}^m$$

form a complete flag. Choose an orthonormal basis $u_1, \dots, u_m$ adapted to this flag. In this basis, $Q_{[t]}$ is diagonal with its first t diagonal entries equal to 1 and the rest equal to 0. Hence

$$M = \sum_{j=1}^n Q_{[t_j]}$$

is diagonal, and its i th diagonal entry is

$$\#\{j : t_j \geq i\} = a_i.$$

Therefore $\det M = \prod_i a_i$, and Proposition 8 gives equality in Theorem 1. This completes the proof.

STATEMENT ON AI USE

GPT-5.4 Pro was used to explore possible proof strategies, test intermediate formulations, and assist with exposition. All mathematical arguments and claims in the final manuscript were independently verified by the author, who takes full responsibility for the paper. The Lean 4 formalization is available from <https://github.com/boonsuan/FerrersBound>.

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